# FrauduLENS: A Credit Card Fraud Detection using Classifier Models and Anomaly Detection

A Technical Report for a Machine Learning
and Data Science Portfolio Project

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2026

**ABSTRACT**

This project investigates the performance of supervised and unsupervised machine learning algorithms in the detection of credit card fraud. It assessed Logistic Regression, K-Nearest Neighbors, and Random Forest classifiers, alongside an Isolation Forest model for anomaly detection. To address class imbalance, we applied the Synthetic Minority Oversampling Technique (SMOTE) and optimized the Random Forest model using Exhaustive Grid Search and Bayesian Optimization. To tackle the problem of class imbalance we applied the Synthetic Minority Oversampling Technique (SMOTE) and for the Random Forest model optimization we applied Exhaustive Grid Search and Bayesian Optimization. Models were tested on Accuracy, Precision, Recall and ROC-AUC scores. The best performance was achieved with the random forest classifier (99–100% accuracy, ROC-AUC 92–98%). The base model was very skewed toward precision (95% precision and 76% recall), but by using SMOTE and hyperparameter tuning, the predictive powers of the model were balanced with a notable 83% precision and 77% recall on the sampled dataset.

## EXPLORATORY DATA ANALYSIS

This section uncovers the explanatory analysis of the Credit Card Fraud Detection dataset from Kaggle using descriptive methods and several data visualization techniques.

## Distribution of Numerical Values

This project utilized a histogram plot to visualize the frequency distribution of each dataset features such as Time, Amount, and V1 to V28 columns which represents the credit card transactions transformed through dimension reduction using Principal Component Analysis. Through this data visualization, we can hypothesize that those V-feature columns with a slightly left or right skewed tail means they are likely to be a case of fraudulent transactions in credit cards, whilst those V-features within the distribution of 0 are likely to be a legitimate one.

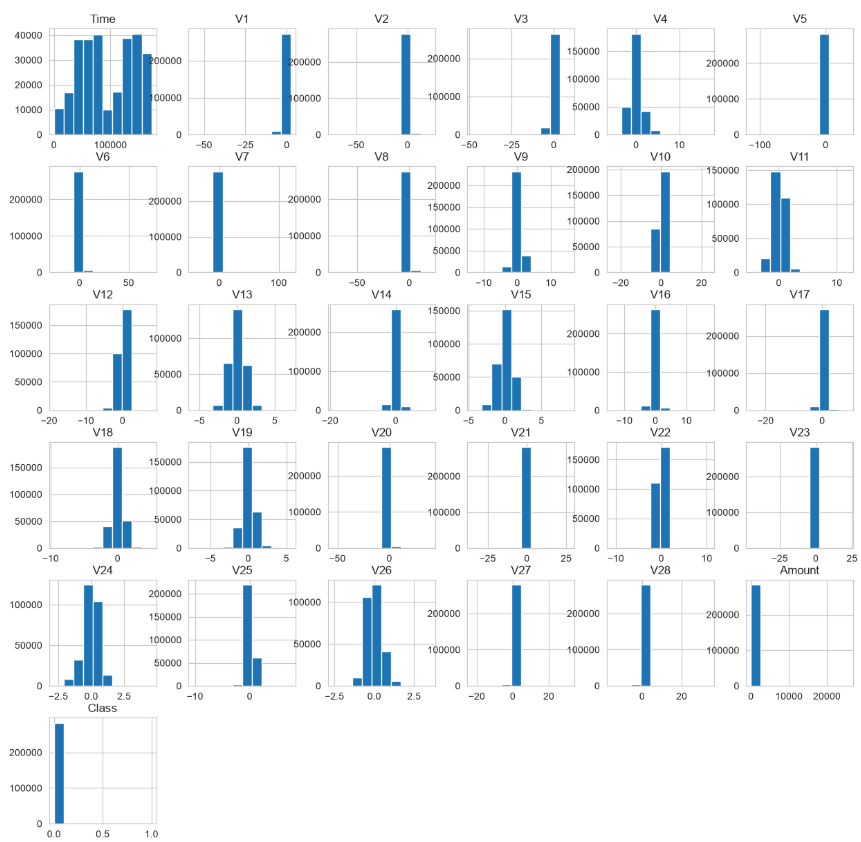


**Figure I.**

**Frequency Distribution**

**Class Feature Counts**

Using a bar chart visualization, we can visualize that there is a significant class imbalance within the dataset. The value representing 0 means that they are legitimate, and 1 are all fraudulent credit cards. The ratio of legitimate credit cards is 99.87%, which is significantly higher than the 0.17% ratio of fraud cases within the dataset. In conclusion, the data demonstrates a highly-imbalanced class count.

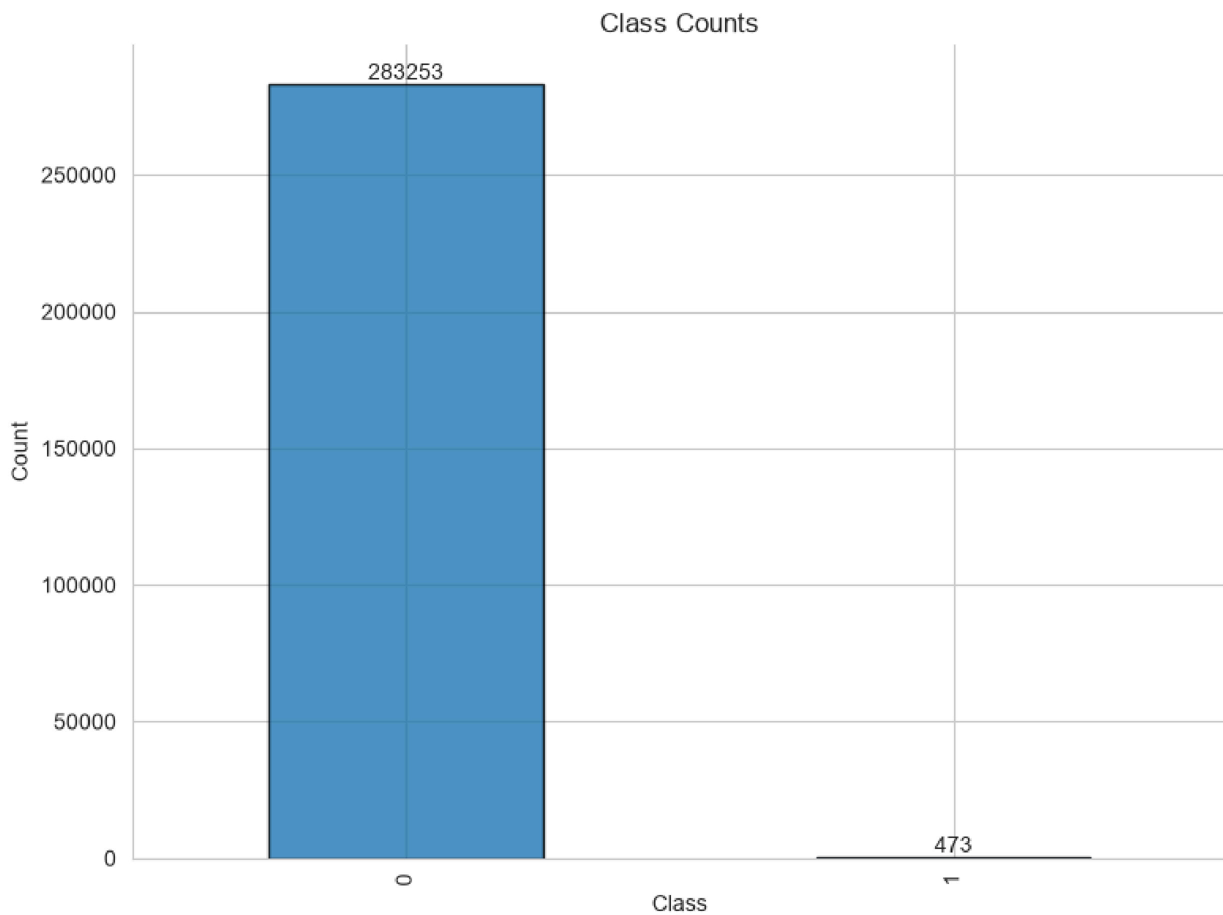


**Figure II.**

**Class Bar Plot**

**Feature Significance**

Using a horizontal bar plot to visualize the feature significance of all the X features against the "Class" target feature. By correlating all the X features against the target feature, we can determine what possible features are significant when we create an initial classifier model. Through the visualization, we can determine that the feature that has the most positive significance for the "Class" target feature is the V11 column, however, the most negative significance is the V17 column feature.

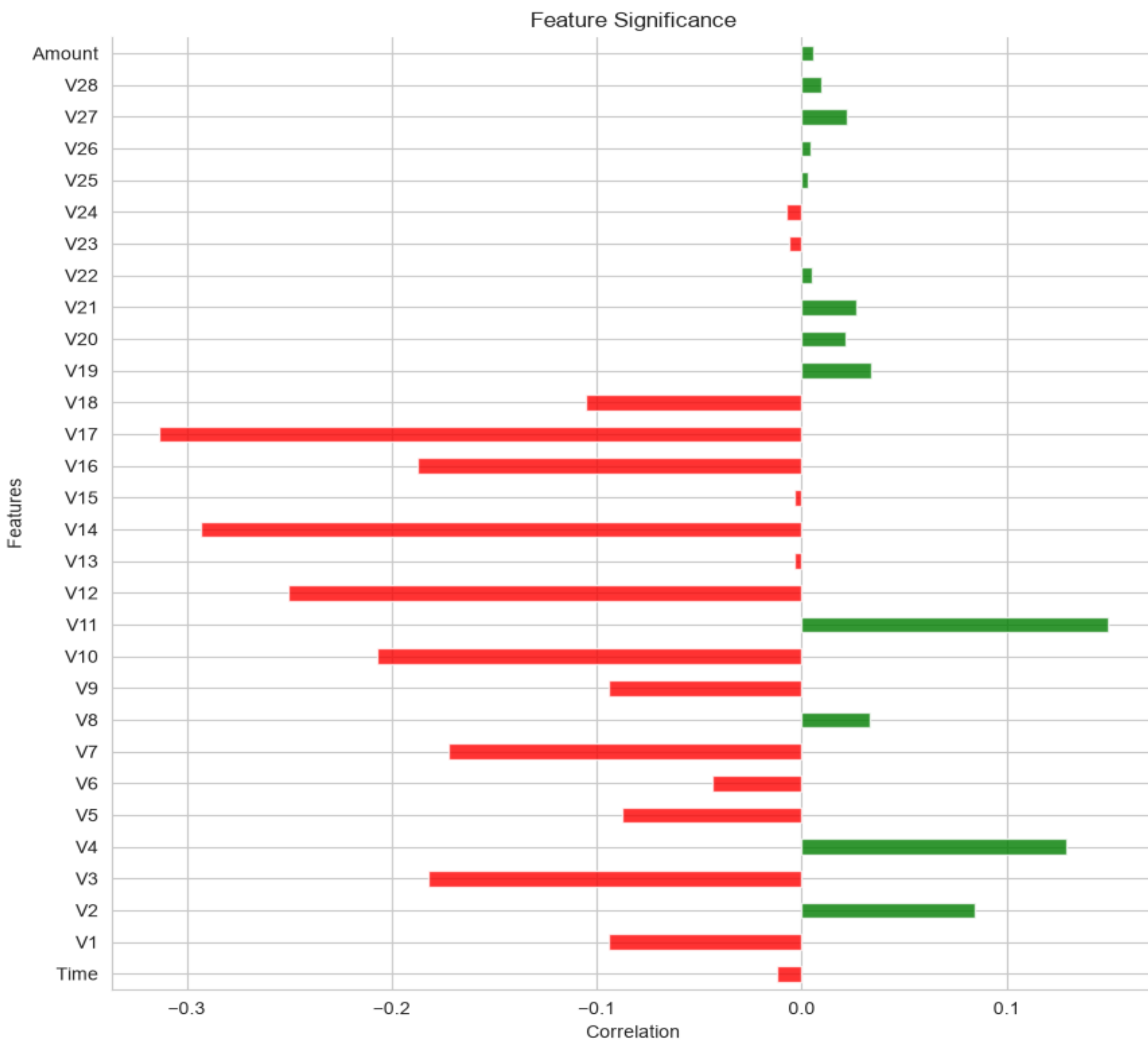


**Figure III.**

**Feature Significance**

## MODEL CREATION

This project modeled several supervised machine learning classifier models such as Logistic Regression as a baseline and linear model, and some complex models such as K-Nearest Neighbors and Ensemble-based Random Forest Classifier. In addition, unsupervised methods such as Isolation Forest are utilized for Anomaly Detection within the dataset. Moreover, this project utilizes several techniques such a SMOTE Sampling for the highly-imbalanced class dataset, and average precision for scoring in hyperparameter tuning to gain a more balanced boundary trade-off between precision and recall.

### Logistic Regression

This baseline model garnered a significant 99% accuracy score, and a 96% ROC-Area Under Curve score of 96%. The following classification report shows the precision and recall trade-off for this model.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 83% | 47% |

**Table I.**

**LR Classification Report**

However, when retraining the model with sampled data using SMOTE sampling, we can see a drop on the accuracy of the model, and a significant drop between the precision and recall trade-off.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 99% |
| **Fraudulent** | 14% | 87% |

**Table II.**

**Sampled LR Classification Report**

**K-Nearest Neighbors**

This model garnered an initial accuracy score of 99%, and a ROC-Area Under the Curve score 90%. On the other hand, the table below shows the classification report between the precision-recall trade-off.

| | Precision | Recall |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 96% | 68% |

**Table III.**

**KNN Classification Report**

On the other hand, by retraining the model using sampled data from SMOTE to address the class imbalance, the model demonstrates just a slight drop on its accuracy, however, it also demonstrates a significant drop on precision-recall trade-off.

| | Precision | Recall |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 45% | 77% |

**Table IV.**

**Sampled KNN Classification Report**

### Random Forests

This model is an ensemble-based classifier model; the trained model garnered a significant 100% accuracy score, with a 92% ROC-Area Under Curve score. The following table below shows the classification report that demonstrates the precision-recall trade-off of the model.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 95% | 76% |

**Table V.**

**RFS Classification Report**

By re-training the model to a SMOTE sampled data to address the significant imbalance class, it shows a more balanced gap and boundary between the precision-recall trade-off garnering 86% for the precision, and 79% for the recall, leaving only 7% gap between the boundary.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 86% | 79% |

**Table VI.**

**Sampled RFS Classification Report**

### **Isolation Forest**

This model is an unsupervised learning model for anomaly detection, this model randomly isolates observations, with shorter path lengths is likely to be considered as anomalies, the following model garnered a ROC-Area Under Curve score of 95%. Using PCA to transform the huge dataset into 2-dimensional components, and using a scatterplot. The visualization was able to detect the anomalies within the dataset, normal data clusters that are likely to be legitimate transactions are within the range between (> 0) and approximately 30. On the other hand, anomalies within the dataset shown are also scattered between 0 to 25 in the X-axis, although these values are scattered all across the axis, with the most outlier lies within (> 175) in the X-axis.

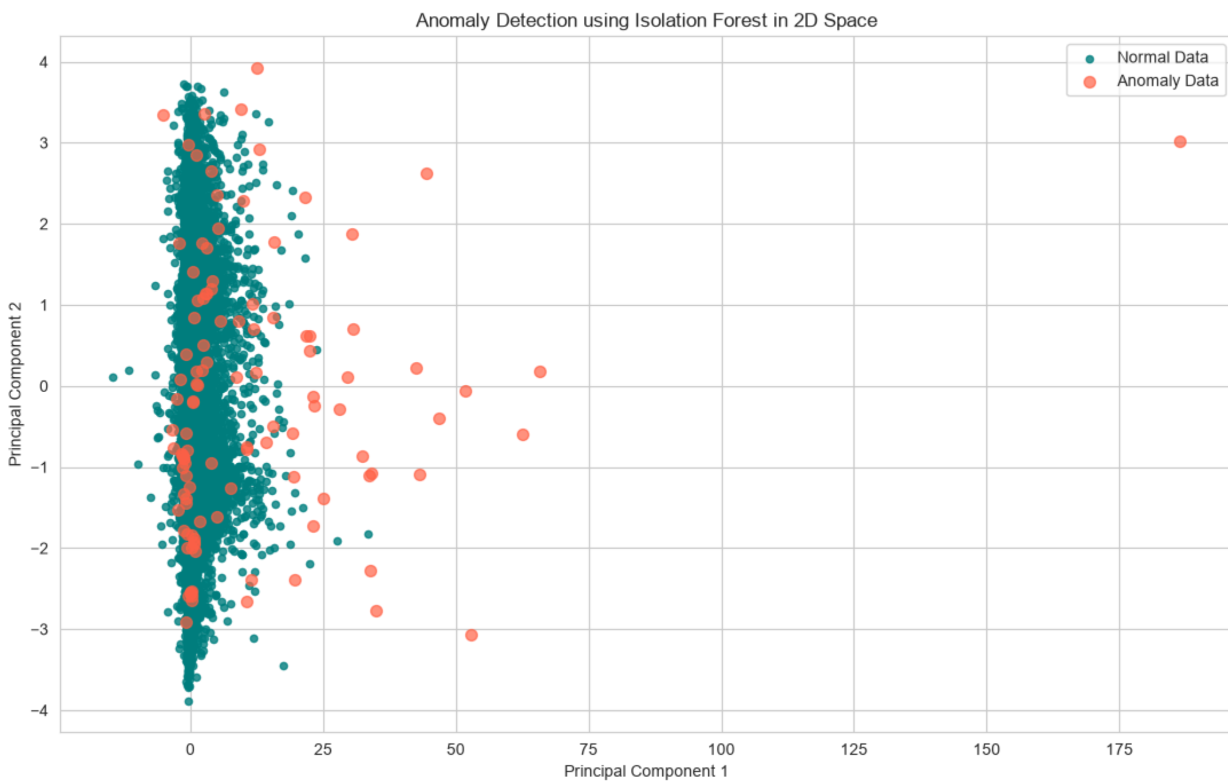


**Figure IV.**

**Anomaly Detection in 2-D Space**

## MODEL EVALUATION

This section aims to demonstrate the model's accuracy and performance, and more importantly, the trade-offs between the precision and recall for each tuned classifier model. Grid Search Cross Validation was used along with Stratified K-Folds with 5 splits for hyperparameter tuning, and searching for the best model estimators and parameters. On the other hand, due to the computational cost of the Grid Search for a more complex model like Random Forests, a Bayesian Optimization Search was used. Lastly, the tuned models were also utilized on sampled data using SMOTE.

### Logistic Regression

The tuned Logistic Regression garnered 97% accuracy score, and a 97% ROC-Area Under Curve score. The tuned parameters used are Lasso regularization, with an inverse regularization score of 100, using a balanced class weight to handle the class imbalance within the dataset, and lastly, using SAGA as an optimization algorithm.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 6% | 89% |

**Table VII.**

**Tuned LR Classification Report**

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 97% |
| **Fraudulent** | 5% | 88% |

**Table VIII.**

**Sampled Data + Tuned LR Classification Report**

**K-Nearest Neighbors**

Another tuned K-Nearest Neighbors model was optimized using Grid Search Cross Validation on the number of neighbors, distance weighting and distance metric (Minkowski power parameter). The optimal set of hyperparameters was found to be 11 neighbors, distance-based weighting and Manhattan distance (p = 1). Using these tuned parameters, the model scored 90% ROC-Area Under Curve. The training accuracy was 100% and test accuracy was 99.95%. The resulting precision recall trade off is shown in the classification report below, with a large increase on the fraudulent class over the untuned model from the previous section.

| | Precision | Recall |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 94% | 74% |

**Table IX.**

**Tuned KNN Classification Report**

To address the class imbalance, the tuned model was retrained on SMOTE-sampled data and the ROC-Area Under Curve score increased slightly to 92%, while the recall for the fraudulent class improved from 72% to 82%. However, this came with a high price to precision, falling from 94% to 33%, following the same precision-recall trade-off pattern seen in the other sampled models across this report.

| | Precision | Recall |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 33% | 82% |

**Table X.**

**SMOTE + Tuned KNN Classification Report**

## Random Forests

For the large ensemble model, due to the computational cost of an exhaustive Grid Search, a Bayesian Optimization Search was used to tune the Random Forest classifier over class weighting, number of estimators, maximum tree depth and maximum features per split. The best hyperparameters were found to be a balanced class weight, a maximum depth of 20, the square-root max-features rule and 299 estimators. The model was able to reach a 99.95% accuracy score and a 97.18% ROC-Area Under Curve score using these tuned parameters, the highest ROC-AUC score among all the tuned models in this report.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 92% | 75% |

**Table XI.**

**Tuned RF Classification Report**

The retraining of the tuned Random Forest on SMOTE-sampled data further improved the ROC-Area Under Curve score (97.68%) and the fraudulent-class recall (77%). The precision was 83%. This SMOTE-sampled tuned Random Forest had the best precision-recall trade-off and the best overall ROC-AUC score of any of the models sampled in this report, and was the best performer of models produced in this project.

| | **Precision** | **Recall** |
|---|---|---|
| **Not Fraudulent** | 100% | 100% |
| **Fraudulent** | 83% | 77% |

**Table XI.**

**SMOTE + Tuned RF Classification Report**

**CONCLUSION**

This project demonstrates the different purpose of machine learning models when demonstrating and predicting fraudulent cases; specifically, in a financial setting through credit card transactions. Furthermore, in a financial setting where a concurrent environment and some important information such as legitimate transactions, customer information, and large sums of money are at stake, pinpointing a high-accuracy model is not enough to consider. It is important to consider efficient decision-making with not just one metric, but also in need to consider a lot of variables.

A Random Forest Classifier stands out among the supervised classifier models as a more consistent model to perform when detecting fraudulent cases in a highly-imbalanced class dataset, while maintaining a consistent and balanced precision-recall trade-offs within unsampled and sampled data. This model minimizes detecting both False Positives and False Negatives without compromising other metrics. In a financial setting, it is essential to minimize errors such as detecting non-fraudulent cases as frauds, and detecting frauds as non-fraudulents.

In this project, it demonstrated that a balanced trade-off between precision and recall is an important metric of this project to minimize the Type I and Type II errors when detecting fraudulent cases or credit cards. In a financial setting, focusing on one metric such as focusing on precision or recall only can compromise one other. A balanced trade-offs ensures a balanced decision making without compromising one another. In addition, using various machine learning models for classifying and detecting anomalies within the dataset helps for a more concrete decision making when it comes to what is an applicable model for a specific task, and by enhancing the model through hyperparameter tuning ensures that the model can achieve their potential use in a specific task.